

SUPERSTORE SALES ANALYSIS

End-to-End Data Analytics Project

Tools Used: Python | Pandas | SQL Server | Power BI | DAX | GitHub

1. Project Overview

The Superstore Sales Analysis project is an end-to-end data analytics project developed to analyze sales, profitability, customer behavior, product performance, regional performance, and discount patterns.

The project starts with a raw Superstore dataset and follows a complete analytics workflow:

```
Raw Dataset
  ↓
Python
Data Cleaning & Exploratory Data Analysis
  ↓
SQL Server
Data Storage & Business Analysis
  ↓
Power BI
Data Modeling, DAX & Interactive Dashboard
  ↓
Business Insights & Recommendations
```

The objective is to demonstrate how raw business data can be transformed into meaningful information that can support business decision-making.

2. Business Problem

A retail business generates large amounts of transactional data containing information about customers, products, orders, sales, discounts, regions and profits.

Simply looking at total sales does not provide enough information to understand business performance.

The business needs answers to questions such as:

- Which products generate the highest sales?
- Which products generate the highest profit?
- Which products are causing losses?
- Which categories perform well?

- Which regions and states generate the most revenue?
- Which regions and states are profitable?
- Which customers contribute the most revenue?
- How does discounting affect profitability?
- How does business performance change over time?

This project addresses these questions using Python, SQL Server and Power BI.

3. Project Objectives

The main objectives of this project are:

1. Clean and prepare the raw sales dataset.
 2. Perform exploratory data analysis using Python.
 3. Store and analyze the cleaned data using SQL Server.
 4. Perform advanced SQL-based business analysis.
 5. Create a proper Power BI data model.
 6. Develop DAX measures for important KPIs.
 7. Build an interactive three-page dashboard.
 8. Identify important business trends and profitability issues.
 9. Provide business insights based on the analysis.
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4. Dataset Description

The project uses the **Sample Superstore** dataset.

The dataset contains transactional information related to customer orders, products, sales, discounts and profits.

Main Columns

Column	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date when the order was placed
Ship Date	Date when the order was shipped
Ship Mode	Shipping method
Customer ID	Unique customer identifier
Customer Name	Customer name
Segment	Customer segment
Country	Country

Column	Description
City	Customer city
State	Customer state
Postal Code	Postal code
Region	Geographic region
Product ID	Product identifier
Category	Product category
Sub-Category	Product sub-category
Product Name	Product name
Sales	Sales revenue
Quantity	Quantity sold
Discount	Discount applied
Profit	Profit generated

The dataset contains approximately **9,994 transaction records**.

5. Python Data Cleaning

Python was used as the first analytical stage of the project.

The main purpose of this stage was to inspect the raw data, identify data-quality issues and prepare a clean dataset for further analysis.

5.1 Data Loading

The dataset was loaded into Python using Pandas.

```
import pandas as pd

df = pd.read_csv("Sample-Superstore.csv")
```

5.2 Data Inspection

The dataset structure was examined using functions such as:

```
df.head()
df.shape
```

```
df.info()  
df.describe()
```

This helped understand:

- Number of records
- Number of columns
- Data types
- Numerical distributions
- Potential missing values

5.3 Missing Value Analysis

Missing values were checked using:

```
df.isnull().sum()
```

The results were reviewed and appropriate cleaning steps were applied where required.

5.4 Duplicate Analysis

Duplicate records were checked using:

```
df.duplicated().sum()
```

This helped ensure that duplicate records did not affect the analysis.

5.5 Data Type Conversion

Date fields were converted into appropriate date formats.

For example:

```
df["Order Date"] = pd.to_datetime(df["Order Date"])  
df["Ship Date"] = pd.to_datetime(df["Ship Date"])
```

Numerical fields were also checked and converted where necessary.

5.6 Column Standardization

Column names were standardized to make them easier to use in SQL Server and Power BI.

For example:

```
Sub-Category → sub_category
```

The cleaned dataset was then exported for SQL Server analysis.

6. Exploratory Data Analysis

After cleaning the dataset, exploratory analysis was performed using Python.

The analysis focused on:

- Sales distribution
- Profit distribution
- Quantity sold
- Discount patterns
- Category performance
- Regional performance
- Product performance
- Customer performance
- Time-based trends

Python libraries such as Pandas, NumPy, Matplotlib and Seaborn were used for analysis and visualization.

The EDA stage helped identify patterns and potential business questions that were investigated further using SQL and Power BI.

7. SQL Server Implementation

The cleaned dataset was imported into SQL Server.

A database named:

```
superstoreproject
```

was created for the project.

The final analysis table used in the project was:

```
dbo.SuperstoreSales
```

The table contains the cleaned transactional data required for analysis.

8. SQL Data Analysis

SQL Server was used to perform structured business analysis.

8.1 Overall KPIs

The following metrics were calculated:

- Total Sales
- Total Profit
- Total Quantity
- Total Orders
- Total Customers
- Profit Margin

Example:

```
SELECT
    SUM(sales) AS Total_Sales,
    SUM(profit) AS Total_Profit,
    SUM(quantity) AS Total_Quantity,
    COUNT(DISTINCT order_id) AS Total_Orders,
    COUNT(DISTINCT customer_id) AS Total_Customers
FROM dbo.SuperstoreSales;
```

9. Category and Sub-Category Analysis

Sales and profit were analyzed at category and sub-category levels.

This helped identify:

- High-revenue categories
- High-profit categories
- Low-profit categories
- Loss-making sub-categories

Technology was identified as an important contributor to both sales and profit.

Furniture generated significant sales but comparatively lower profit.

Tables were identified as a major loss-making sub-category.

10. Regional and State Analysis

The dataset was analyzed by:

- Region
- State
- City

Regional analysis was performed to identify differences in sales and profitability.

The West region generated the highest overall sales and profit in the analysis.

State-level analysis also revealed that some states can generate substantial sales while still producing negative profit.

For example, Texas recorded high sales but negative overall profit.

This demonstrates why sales should not be evaluated independently from profitability.

11. Product Analysis

Products were ranked according to:

- Sales
- Profit

SQL ranking functions were used to identify top and bottom-performing products.

The analysis included:

- Top 10 products by sales
- Top 10 products by profit
- Bottom 10 products by profit
- Product ranking within categories
- Top 3 products in each category

Window functions such as `RANK()` and `PARTITION BY` were used for advanced product analysis.

12. Customer Analysis

Customer performance was evaluated using:

- Total Sales
- Total Profit
- Total Orders

Customers were ranked using SQL window functions.

The analysis also compared sales and profitability to identify whether high-sales customers were also highly profitable.

This is important because a customer can generate significant revenue while producing comparatively low profit.

13. Time-Based Analysis

Sales performance was analyzed by:

- Month
- Year
- Year-Month

SQL window functions such as `LAG()` were used to calculate month-over-month changes.

A running total was also calculated to understand cumulative sales performance.

These analyses help identify business growth patterns and changes in performance over time.

14. Discount and Profit Analysis

Discount levels were analyzed against profitability.

Transactions and products were examined based on discount levels and profit.

The analysis indicates that higher discounts are generally associated with weaker profitability in several areas.

However, this relationship should be interpreted as an association rather than proof that discounts directly cause lower profits.

15. Advanced SQL Techniques

The project demonstrates several advanced SQL concepts:

Aggregation

```
SUM()  
COUNT()  
AVG()
```

Conditional Logic

```
CASE  
  WHEN ...  
  THEN ...  
END
```

Common Table Expressions

```
WITH CTE AS (...)
```

Ranking

```
RANK()  
DENSE_RANK()
```

Window Functions

```
LAG()  
OVER()  
PARTITION BY
```

These techniques were used to solve practical business questions rather than being included only as technical demonstrations.

16. Power BI Data Model

The cleaned SQL Server table was connected to Power BI using Import mode.

The primary fact table is:

```
SuperstoreSales
```

A separate date table was created using DAX.

```

DateTable =
ADDCOLUMNS(
    CALENDAR(
        MIN(SuperstoreSales[order_date]),
        MAX(SuperstoreSales[order_date])
    ),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month", FORMAT([Date], "MMM"),
    "Year Month", FORMAT([Date], "YYYY-MM"),
    "Quarter", "Q" & FORMAT([Date], "Q")
)

```

A one-to-many relationship was created:

```

DateTable[Date] 1 : * SuperstoreSales[order_date]

```

This date model allows time-based analysis throughout the dashboard.

17. DAX Measures

The following measures were created in Power BI.

Total Sales

```

Total Sales =
SUM(SuperstoreSales[sales])

```

Total Profit

```

Total Profit =
SUM(SuperstoreSales[profit])

```

Total Quantity

```

Total Quantity =
SUM(SuperstoreSales[quantity])

```

Total Orders

```
Total Orders =  
DISTINCTCOUNT(SuperstoreSales[order_id])
```

Total Customers

```
Total Customers =  
DISTINCTCOUNT(SuperstoreSales[customer_id])
```

Profit Margin

```
Profit Margin =  
DIVIDE(  
    [Total Profit],  
    [Total Sales],  
    0  
)
```

Average Order Value

```
Average Order Value =  
DIVIDE(  
    [Total Sales],  
    [Total Orders],  
    0  
)
```

Average Discount

```
Average Discount =  
AVERAGE(SuperstoreSales[discount])
```

18. Power BI Dashboard

The Power BI report consists of three analytical pages.

Page 1 – Executive Sales Dashboard

Purpose

Provides a high-level overview of business performance.

KPIs

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Profit Margin

Visualizations

- Monthly Sales & Profit Trend
- Sales by Category
- Profit by Category
- Sales by Region
- Top 10 Products by Sales

Slicers

- Year
- Region
- Category
- Sub-Category
- Segment

A Reset button was also added to return the page filters to their default state.

19. Page 2 – Product & Customer Analysis

Purpose

Analyzes product and customer performance.

KPIs

- Total Sales
- Average Order Value
- Average Sales per Customer

Visualizations

- Top 10 Products by Sales
- Top 10 Products by Profit
- Bottom 10 Products by Profit
- Customer Sales vs Profit

- Customer Performance Table
- Category → Sub-category → Product Matrix

Filters

- Year
- Region
- Category
- Segment
- Product Name

The page includes a Reset button for returning filters to the default state.

20. Page 3 – Profitability & Regional Analysis

Purpose

Focuses on profitability and geographical performance.

KPIs

- Total Sales
- Total Profit
- Profit Margin
- Average Discount

Visualizations

- Sales & Profit by Region
- Profit Margin by Category
- Discount vs Profit
- Profit by Sub-category
- State Performance
- Loss-making analysis

Filters

- Year
- Region
- Category
- Sub-category
- Segment

A Reset button was added for easier dashboard interaction.

21. Key Business Insights

Insight 1 – Technology is a major business driver

Technology generates approximately **\$836K in sales** and approximately **\$145K in profit**.

This makes Technology an important contributor to both revenue and profitability.

Insight 2 – Furniture has a profitability gap

Furniture generates approximately **\$742K in sales**, but its profit is only around **\$18K**.

This indicates that high sales volume does not necessarily result in strong profitability.

Insight 3 – Tables are loss-making

The Tables sub-category records approximately **-\$17.7K in profit**.

This makes Tables an important area for further investigation.

Insight 4 – West is the strongest region

The West region generates approximately **\$725K in sales** and around **\$108K in profit**.

It is a significant contributor to overall business performance.

Insight 5 – Some high-sales states are still unprofitable

Texas generates significant sales but records approximately **-\$25.7K in profit**.

This demonstrates the importance of monitoring profit alongside revenue.

Insight 6 – Discounts require monitoring

The analysis indicates an association between higher discounts and weaker profitability in several parts of the business.

Discount policies should therefore be evaluated alongside product margins and customer demand.

Insight 7 – Customer sales and profitability should be evaluated together

A customer generating high sales is not necessarily generating high profit.

Therefore, customer evaluation should consider both revenue and profit.

22. Business Recommendations

Based on the analysis, the following areas can be investigated further:

1. Review loss-making products

Analyze products with negative profit to understand whether pricing, discounts or costs are responsible.

2. Review the Tables sub-category

Investigate pricing, discount levels and product-level profitability before increasing sales volume.

3. Analyze high-sales but low-profit states

States such as Texas should be investigated to understand why high revenue does not translate into positive profit.

4. Improve discount strategy

Discounts should be monitored carefully, especially for products that already have low margins.

5. Focus on profitable products

High-profit products can be prioritized for inventory planning and promotional strategies.

6. Monitor customer profitability

Customer retention strategies can consider both sales value and profit contribution.

23. Project Outcome

This project demonstrates a complete data analytics workflow from raw data to business insights.

The project successfully combines:

`Python`
↓

Data Cleaning & EDA

SQL Server

↓

Data Storage & Business Analysis

Power BI

↓

Data Modeling + DAX + Visualization

Business Insights

↓

Decision Support

The project demonstrates practical skills in:

- Data Cleaning
- Exploratory Data Analysis
- Python
- Pandas
- SQL
- Advanced SQL
- Data Modeling
- DAX
- Power BI
- Business Intelligence
- Dashboard Development
- Business Analysis

24. Conclusion

The Superstore Sales Analysis project shows how data can be transformed from raw transactional records into useful business information.

Python was used to prepare and explore the data, SQL Server was used for structured and advanced business analysis, and Power BI was used to communicate the findings through an interactive dashboard.

The analysis highlights that business performance should not be measured using sales alone. Profitability, discounts, products, customers, regions and states must also be considered to understand the overall health of the business.

The final dashboard provides an interactive way to explore these factors and supports data-driven business analysis.

25. Author

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